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Risk factors for rheumatoid arthritis

Kimmo Aho and Markku Heliövaara

Both genetic and environmental factors contribute to the development of rheumatoid arthritis (RA). A long-term latent process often precedes the onset of arthritis. Hence, the ultimate causes of RA cannot be clarified by studying only the inflamed joints. Longitudinal studies focusing on risk factors are crucial in approaching the true aetiology. At present, most information gained from epidemiological studies is contradictory or vague. For instance, there is no consensus concerning the long-term effects of pregnancy or the putative protective role of oral contraceptives. There is no doubt that diet plays a role, but no specific nutrient has proved to be either protective or deleterious. Smoking is the only environmental risk factor that has been firmly verified epidemiologically for RA. It can be reasonably regarded as a contributory cause of RA. Studying the immunological effects of exposure to cigarette smoke may offer an opportunity to combine information from basic and epidemiological research to clarify the causal chains leading to RA.

Keywords: dietary factors; epidemiology; genetics; reproductive factors; rheumatoid factor; smoking

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Introduction

Rheumatoid arthritis (RA) is a chronic, systemic inflammatory disease. It is the most important of all forms of arthritis. Genetically related and environmentally associated determinants interact over time, contributing to the disease. Immune-mediated mechanisms certainly play a role in the pathogenesis of RA, but we are still ignorant concerning the basic mechanisms that initiate and sustain it. There seems to be no single determinant either sufficient or necessary for its causation. Here we review the available informa-

tion on genetic and environmental risk factors for RA and attempt to relate these to the available knowledge on the pathogenesis of the disease.

Genetic versus environmental factors

Twin studies provide a classic approach for separating the influence of genetic and environmental risk factors in the disease: a higher concordance rate in monozygotic compared to dizygotic twins is consistent with inherited determinants. However, the absolute levels of concordance and the ratio between the concordance rates are dependent on the prevalence of the disease (1, 2). The above ratio is also related to the number of genes involved (1). The monozygotic concordances of 12% (3) and 15% (4) observed in two population-based surveys of RA twins are often erroneously taken to mean that the genetic component of RA is fairly small.

A more readily interpretable measure for evaluating the significance of genetic factors is provided by computing 'heritability', which estimates the extent to which variation in the susceptibility to the disease in a population can be explained by genetic variation and it is independent of disease prevalence. Heritability was assessed by variance component analysis using data from the two twin series mentioned above, one from Finland (3) and the other from the United Kingdom (4), and proved to be about 60% (2). Although tempered by power considerations, there was no evidence in these twin data that the overall genetic component of RA would differ for sex, age, age at disease onset, or severity of disease.

Heritability estimates, however, tell only about the prevailing individual genetic differences and cannot, in general, be extrapolated to other populations and other environments (5). For example, twin studies indicate that the heritability of human stature is high and environmental variation is quite small. However, a marked increase in human stature that has almost certainly resulted from changes in the environment has taken place during the past two centuries in

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populations of European ancestry. Thus, large environmental effects may be observed even where heritability measurements apparently indicate only a small environmental component. A real advance in genetic knowledge will occur only when we are able to analyse the existing genetic variation in depth by isolating single genetic components and then show how the effects of the genes are mediated.

Family studies

The degree of familial clustering is probably the best way to measure the genetic component of the disease. The clustering can be estimated from the prevalence of disease between two comparison populations, siblings of affected individuals and controls (λ_s) usually consisting of the general population. In early studies, a wide variation had been noted in the λ_s values (6), resulting either from flaws in study design or from chance variation due to a small number of cases. Several studies had relied on the family history as elicited by the probands, and it is possible that RA patients were more aware of the occurrence of RA in their relatives than were the controls. In addition, most of the studies did not adjust for age and gender difference between the family members of the probands and the controls. Indeed, a critical analysis of the early studies concluded that the available data gave little support for any genetic cause of RA (7).

On the basis of more recent and better designed studies, it is now firmly established that there is an increased rate of occurrence of RA in close relatives of RA probands, although the sibling recurrence risk is appreciably lower than in type I diabetes or in systemic lupus erythematosus, for example. Thus, λ_s values ranging from 1.6 to 4.4 have been reported during the past two decades (8–11). Figures from population-based series have been lower than those from hospital-based series and the family members of probands with an early age at presentation have shown a higher prevalence of RA than family members of probands who have fallen ill at an older age. In the Finnish nationwide twin series, the age- and gender-adjusted increase in RA prevalence among monozygotic co-twins was 8.6-fold and among dizygotic co-twins 3.4-fold compared with the remaining twin population (3). Finally, substantially increased incidence of RA was noted in previously unaffected members in families in which there were already multiple cases of RA (12).

The role of HLA

RA is known to be associated with human leukocyte antigen (HLA) class II alleles, particularly with

Key messages

- RF-positive and RF-negative RA have different risk factors
- Smoking is the only environmental risk factor that has been firmly verified for RA

certain subtypes of HLA-DRB1*04 (HLA-DR4). Indeed, the initial discovery of the association between HLA-DR4 and RA (13) has been replicated in many populations (6). Associations with other HLA-DRB1 specificities, such as DRB1*01 and DRB1*10, have also been described. All the DRB1 alleles associated with RA share a very similar amino-acid sequence, i.e., the 'shared epitope', that is located in the third hyper-variable region coded by that gene (14). Subjects who are 'homozygotic' for the shared epitope are at a greater risk of developing RA than those carrying only a single copy. Furthermore, there appear to be differences in the strength of association depending on different alleles and allele combinations. Certain rare combinations seem to carry quite a high risk. Although the great majority of the RA patients in populations of North European ancestry are positive for the shared epitope, in certain other populations (frequently with low prevalence of RA and HLA-DR4) a substantial proportion of RA patients are negative for it. In such populations, there may be an insufficient density of the susceptibility epitope to override or dominate environmental influences (15).

The location of RA-associated genes in the HLA-D region, known to be analogous to the region encoding immune response genes in mice and monkeys, allowed speculations as to the role of the immune system in RA. However despite intensive research during the 25-year period since the initial observation of the HLA association (13), very few hard facts have come out concerning the mechanisms with which the effects of HLA-DRB1 genes are mediated. Most of the evidence points out that the HLA alleles have a greater impact on disease severity than on susceptibility to it, although there are studies that are not in line with this concept (6). Besides association with the shared epitope, certain other genes in the HLA region have been connected with RA. One explanation is linkage disequilibrium with DRB1, although there is evidence that at least some associations would be independent of the shared epitope (16, 17).

The contribution of the HLA region to genetic susceptibility to the disease (λ_s for HLA) can be estimated on the basis of haplotype sharing in affected sib pairs by computing the ratio of the expected proportion of sib pairs sharing zero alleles to that

observed (18). Three large sibling series have shown similar figures to each other, ranging from 1.6 to 1.8 (19–21). Provided that the overall sibling recurrence risk is as low as two, most of the susceptibility to RA would be encoded by HLA: if the recurrence risk were of an order of magnitude of five, other genes would encode about two-thirds of the susceptibility. In the English twin series, ‘homozygosity’ for the shared epitope was the most important factor in determining RA concordance in monozygotic twins; RA concordance was low in those twins who were negative for the shared epitope (22). This suggests (although the figures are small) that in the absence of the shared epitope only a relatively small genetic effect remains for RA.

Linkage to HLA has been observed in all major genome-wide screens of linkage (23). No other specific genes have been identified with certainty, although it has been suggested that several different loci are involved in genetic susceptibility. This would mean that the remaining genetic risk is either distributed among several different loci, with each locus separately contributing a much lower degree of risk than HLA does, or that it is so small that there are no chances whatsoever of finding anything meaningful. Time will tell whether the considerable investment of time and money will lead to identification of additional susceptibility loci.

Rheumatoid factor

RA is associated with several autoantibodies specific enough to serve as diagnostic and prognostic markers (24–26). Of these, the most attention has been paid to the rheumatoid factor (RF). It is an autoantibody that binds the Fc region of the immunoglobulin G (IgG) molecule (27, 28). RF can be induced by cell mitogens and by immunization with a variety of different antigens. Induction by the antigen-driven route has been accomplished only in secondary but not in primary immunisation, suggesting that RF arises as a response by antigen/antibody-complex rather than the antigen alone.

Sustained RF production in infections and immunizations depends on continued immunological stimulation. Elimination of the immunizing agent leads to a rapid decline of the RF titre. On the other hand, RF is produced in large amounts in RA patients and often over a whole life-time. It is commonly believed that immune complexes containing RF constitute a self-perpetuating system amplifying the inflammatory process by complement activation and/or by interaction with monocytes. The initiating factor, however, has been enigmatic, since no exogenous antigen has been found in complexes.

There are healthy subjects with positive RF

reactions for which there is no obvious explanation. In a considerable proportion of such subjects the reactions tend to persist for many years, if not their whole life (29, 30). Thus, the mechanisms leading to RF production among them are likely to be different from those in infections and immunizations. Some of the healthy subjects will develop RA in due course (31–33). Thus, RF can be regarded as a risk factor for RA. Furthermore, determinants of RF positivity at the community level may be aetiologically related to RA.

Although RA is associated with HLA-DR4, three fairly large series do show that there is no association whatsoever with ‘false-positive’ RF and HLA-DR4 (34–36). This would indicate a selective process among RF-positive subjects concerning the development of RA.

Virtually nothing is known about the population associations of other marker antibodies of RA. A statistically significant association was observed between RF and antifilaggrin antibodies in sera of healthy subjects who later developed RA and in sera of control subjects, indicating that these two antibodies are associated markers of the rheumatoid immunological process (37).

Seropositive versus seronegative arthritis

For any epidemiological study to proceed, the disease ought to have a clear definition. There are considerable problems concerning RA. Several sets of criteria have been proposed, of which the most recent one was from the American College of Rheumatology in 1987 (6). A basic problem with RA is that there is no single criterion that applies exclusively to it. The principal purpose of classification criteria is to facilitate accurate communication (38). Classification criteria may also be used to define cases for epidemiological research. Because treatment decisions are not made on the basis of whether subjects satisfy a case definition, some ‘misclassification’ is acceptable.

About one-third of the patients initially presenting with symptoms and signs compatible with RA are seronegative according to the results of conventional RF tests. RF-negative RA most likely consists of a heterogeneous group of disease conditions. For example, the patient may have HLA-B27-associated peripheral arthritis without evidence of spinal involvement, hidden psoriatic arthritis without overt skin lesions, or juvenile idiopathic arthritis of adult onset. Depending on the cut-off level of RF positivity, some RF-positive cases will remain in the group of patients classified as RF negative.

On average, RF-negative patients tend to have a more favourable course than those with positive reactions, and many go on to complete remission.

Furthermore, patients with RF-positive RA have a significantly higher frequency of extra-articular manifestations, such as subcutaneous nodules, rheumatoid pulmonary disease and vasculitis. A variable but relatively small proportion of RF-negative patients who develop chronic progressive disease that can be resistant to conventional therapies remain however.

In the search for risk factors for RA, it may be better to deal separately with RF-positive and RF-negative chronic peripheral arthritis than to lump together those cases that satisfy the criteria of the American College of Rheumatology classification.

When does rheumatoid disease start?

There is increasing evidence to indicate that at least some key facets of the disease process in RF-positive RA begin prior to the onset of clinically manifest arthritis. Nothing is known concerning RF-negative RA.

The available histopathological studies of clinically uninvolved joints have demonstrated the presence of subclinical synovitis in affected individuals (39–41). Moreover, comparison of synovial tissue histopathology between patients with early RA and RA of longer duration revealed obvious similarities (42). These findings can be taken to imply that there is a subclinical phase in RA in which synovium is abnormal without actually being clinically inflamed and that 'early RA' is already a chronic disease. The length of this subclinical phase is not known.

As already mentioned, positive RF frequently precedes the onset of clinically manifest RA (31–33, 43). The same is true for filaggrin-related antibodies (37, 43, 44). The prevalence of positive reactions is dependent on the period between taking the specimen and the onset of the disease. When the period is long (over 10 years) the prevalence is fairly low, whereas when it is short (less than 2 years) the prevalence is nearly the same as in the established disease (24). Increased IgG levels antedated the clinical RA even for a longer period than the above autoantibodies (45).

As discussed later, smoking has been established as a risk factor for RF-positive (but not for RF-negative) RA. Importantly, the increased risk for RF-positive RA remained for somewhere up to 15 to 20 years after smoking cessation (46, 47). Thus, smoking has long-lasting effects, which are crucial in the development of RA.

The pre-onset natural history of (RF-positive) RA can broadly be taken to consist of induction period, latent period, and events leading to manifest disease. Some determinants may exert their influence by worsening existing symptoms. It is entirely possible that RA has some of its roots in infancy. Thus, there

is some evidence that fetal adaptations to maternal and environmental influences programme events in later life (48). Such programming may play a role for instance in coronary heart disease and in type II diabetes. The possibility that the same might be true for RA cannot be excluded.

Role of infections

Considerable attention has been focused on the infectious aetiology of RA (49). It is well established that intercurrent infections are often associated with flare-ups of RA. Accordingly, it is not unrealistic to assume that infections and immunisations could in some instances act as triggering agents for RA in subjects with genetic and immunological susceptibility. Indeed, there is some evidence for the role of vaccinations (50, 51). Interpretation of the findings, however, is not always straightforward; vaccination might only be a surrogate marker for some other factor such as trauma.

There are several different mechanisms by which infection could play a role in the earlier phases of RA, for instance by contributing to the development of autoimmunity through mechanisms such as molecular mimicry. However, because the interval between the putative initiating infection and the onset of clinical RA may be quite long, it is difficult to demonstrate the role of infection.

A recent approach has been to use polymerase chain reaction techniques to identify microbial DNA in joints. However, the absence of DNA does not rule out the role of microbes in pathogenesis since the organism might be in the lesion in the earliest phase only. On the other hand, inflamed joints have greater accessibility to circulating micro-organisms and, accordingly, the recovery of any organism from such joints may have no relevance to disease pathogenesis. Indeed, findings concerning RA have been difficult to interpret or have not been replicated (52, 53).

Descriptive epidemiology

The first stage in an epidemiological study of a disease is to describe the manner in which it appears in a population, i.e., to describe levels of disease frequency, geographic distributions, patterns according to various demographic features, and trends over time. Numerous reports have provided such information for RA from many parts of the world.

Several recent studies have provided fairly accurate estimates on the influence of age and gender on the incidence of RA (54–57). The overall incidence of this disease is two to three times higher in women than in men. The incidence increases up to the age group 65–

74 years, whereafter there seems to be a small decline. In prevalence studies, the decline is more marked due to excess mortality associated with RA. According to all the above studies, RA seems to be fairly rare in men under 35. Thereafter, the gender difference tends to narrow so that there is only a slight excess of women in the oldest age groups.

Earlier, the prevalence data available were interpreted as showing that the frequency of occurrence of RA was about the same throughout the world. Indeed, prevalence between 0.5% and 1% has been reported in several European and North American populations (6). Nevertheless, somewhat lower figures have been noted in some Mediterranean countries, and South-East Asia and also in China and Japan. On the other hand, figures as high as 5% have been recorded in some native American-Indian populations, whereas RA seems to be rare in some tropical areas. Both genetic and environmental factors may be involved here. According to one study, Pakistanis living in the UK had a higher prevalence than those living in Pakistan, but not as high as the prevalence in the ethnic British population (58). Recently, regional differences in the incidence of RA were observed in the genetically homogenous Finnish population (59).

Changes over time in the occurrence of RA in its age and sex distribution might imply that specific environmental factors have had an effect. A clear-cut decline in the incidence of RA has taken place in Pima Indians (60) and in the population of Rochester, Minnesota (61). On the other hand, there is some evidence of an increase in the occurrence of RA in Sub-Saharan Africa, possibly related to urbanisation (62). A shift in the incidence of RA toward elderly patients has been noted in Finland (55) and in Japan (63). Increased life expectancy might have had some influence on this; however, a marked decline in the incidence of RA in younger age groups was noted in both studies.

Reproductive factors

Many studies have documented the remission in symptoms of RA during pregnancy since the first study by Hench in 1938 (64). However, activity of the disease tends to recur during the postpartum period. A similar type of relationship seems to affect the incidence of new cases; there is some reduction in the onset of RA during pregnancy and an increased risk of development of RA symptoms during the postpartum period (65, 66). Therefore, it seems as if pregnancy somehow postpones the outbreak of manifest disease. According to one case-control study, much of the increased RA risk during the postpartum period seemed to be associated with lactation (67). A

possible explanation for the association with breastfeeding is its relationship with prolactin, a proinflammatory hormone (68).

Until recently, relatively little has been known about the long-term effects of pregnancy. A number of studies from 1990s suggested that women who are nulliparous have an increased risk of developing RA (6), whereas two large population-based cohorts failed to show any effect (69, 70).

An unexpected finding from the Royal College of General Practitioners study of the effects of oral contraceptives on various aspects of women's health found that incidence of RA among users of oral contraceptives was reduced by 50% (71). A subsequent follow-up of the same cohort showed that this gap in incidence narrowed, suggesting that oral contraceptive use postpones rather than prevents the development of RA (72). Two further cohort studies did not reveal any significant effect, whereas most, but not all case-control studies have shown some reduction in the incidence (6). In one case-control study in which a clear reduction in the incidence was noted, no difference emerged between RF-positive and RF-negative cases (73). It has to be noted that the composition of contraceptive pills has varied between countries and over time. As a whole, the available data do not exclude a modest protective effect whilst using oral contraceptives.

Smoking

Among the risk factors for diseases, smoking represents the widest spectrum of outcomes and the heaviest burden in terms of disability and mortality. Since information on smoking is therefore essential in many clinical and epidemiological studies, it is surprising that the close associations of smoking with RF and RA were not realised until the 1990s.

The association between smoking history and the prevalence of false positive RF was first observed in the Mini-Finland Health Survey (74) and subsequently replicated in Iceland (75). The association prevails in both sexes and is strong enough to have biological significance. The higher the RF titre, the more closely it is associated with smoking, but no dose-response gradient has been observed for intensity or duration of smoking (74, 76, 77). The association may be irreversible: ex-smokers continue to produce RF after quitting (74, 76). However, no prospective study has been focused on smoking as a predictor of RF.

There is accumulating evidence that smoking is a risk factor for RF-positive RA (or any RA) but not for RF-negative RA (46, 47, 76, 78–85). In Finland and Norway, the association between smoking and RA is stronger in men than in women (46, 83, 86), whereas

such interaction does not seem to prevail in Sweden (47). Among women, the association is stronger after menopause; this has led to speculation that hormonal factors could play a role in such interaction (85, 86). In a British case-control study of both sexes, the association between smoking and RA was found to be stronger among those without a family history of RA, suggesting interplay with genetic factors (87). In all, the RA risk of current cigarette smokers compared with those who have never smoked seems to be between 2 and 4 among men and 1.5 and 3 among women.

The risk of developing RA seems to occur above a certain threshold of about five to ten cigarettes each day for about 20 to 30 years. No dose-response relation is evident at higher consumption levels (47). In line with studies focusing on RF, the relative risk remains high for at least 15 years after cessation of smoking (46, 47). A dose-response gradient, however, has been suggested in three studies (82, 84, 87). Such a monotonic relation between exposure and outcome supports causality, but its absence is not falsifying.

In line with the association between smoking history and RF, smoking also seems to influence the severity of RA (88, 89). In a study of female RA patients, the association between current smoking and RF status was significant only in glutathione S-transferase M1-null patients (90). On the basis of this suggestive finding, the authors speculated that a lack of that enzyme in smokers might promote RF production through a failure to detoxify smoke-derived chemicals or their by-products, which have the potential to damage IgG (90). Whatever the mechanism, such interaction between a gene and smoking history adds to the evidence that smoking is a causal risk factor for RA, in part through RF production. However, nothing is known about the role of glutathione S-transferase enzyme in subjects with 'false-positive' RF.

None of the recent reviews has inferred that smoking is a causal risk factor for RA (6, 91–93). However, it may be time to regard smoking as a causal risk factor for RF-positive RA, although the mechanism is unknown. Smoking represents a risk factor evidently contributing to the induction, i.e., to RF preceding RA. In future studies, the interaction of smoking with age, sex, hormonal factors and genes may provide clues for an understanding of the aetiology of RA.

Coffee

Three prospective studies have reported associations between coffee consumption and the risk of RA independent of smoking. The first study focused on Scandinavian-type boiled coffee only, and it also

observed a cross-sectional association between coffee consumption and the prevalence of RF in the absence of arthritis (75), whereas in the latter studies only decaffeinated coffee predicted the incidence of RA (94, 95). In contrast, a recent cohort study of women revealed no significant association between consumption of coffee, decaffeinated coffee, or total coffee and risk of developing RA (96).

Diet

There has traditionally been a strong belief that specific diets influence the course of RA. However, it was not until the 1990s that cross-sectional or retrospective epidemiological studies focused on dietary factors as potential risk factors. It was suggested that diets rich in fish (97–99), olive oil (98, 100) and cooked vegetables (100) provide protection against RA. In accordance with the *ad hoc* epidemiological studies, positive associations between serum cholesterol and RF (76) and, in males, RF-positive RA (101) have been suggested. A recent randomised trial indicated the favourable effects of a Mediterranean diet on inflammatory activity and physical function in RA patients (102). High intake of n-3 fatty acids from fish and oleic acid from olive oil produce anti-inflammatory effects, which could explain the observations between diet and RA risk (102). But such diets are also rich in natural antioxidants.

Free oxygen radicals have been implicated as mediators of tissue damage in RA. Antioxidant micronutrients may therefore play an important role as part of the mechanism that protects against tissue damage in RA. This serves as a hypothesis for observational studies. In line with the hypothesis, in two nested case-control studies using stored serum samples, selenium, alpha-tocopherol, beta-carotene, and an overall antioxidant index were inversely associated with the later development of RA (103–105). In a case-control study assessing dietary intake using a food frequency questionnaire, intake of vitamin C, but not of vitamin E, was inversely associated with the risk of RA (99). In the Iowa Women's Health Study, the intake of beta-cryptoxanthin predicted a reduced incidence of RA, while no associations were found with other carotenoids (106). In that study, supplemental use of zinc, copper and manganese, and intakes of fruits and cruciferous vegetables also showed inverse associations with RA risk. In the cohort of the Finnish Mobile Clinic Health Examination Survey, however, higher intakes of two dietary flavonoids, kaempferol and quercetin, predicted an increased incidence of RA (107). The associations were mainly related to RF-positive RA. Of the dietary sources rich in flavonoids, intake of white cabbage was strongly associated with the risk

of RF-positive RA. In all, the evidence is insufficient to conclude that some definite antioxidant, or intake of antioxidants in general, would provide protection against RA.

Measuring dietary intake in epidemiological studies remains a problem. Another major concern is confounding; the inability to distinguish effects of various nutrients from each other, or the overall effect of diet from that of other lifestyle factors, detracts from the validity of the diet-RA associations observed. To sum up, dietary factors certainly affect the risk of RA, but the identification of crucial nutrients may not be possible through epidemiological studies.

Other factors

Since atopic disorders and RA present opposite immunological cytokine profiles, atopy has been hypothesised as providing protection against RA. The results of two studies support the hypothesis by showing a negative co-morbidity (108, 109). Studies from Finland and Sweden have found an increased risk of RA among workers exposed to silica (110, 111), whereas there was no evidence on such an association in a study from the UK (112). In a case-control study, a strongly positive association was found between RA and a history of blood transfusion (81), whereas an extensive cohort study of middle-aged women showed an inverse association (113).

Concluding remarks

There are many studies on risk factors for RA; these

show contradictory or unconfirmed results. As assessed retrospectively, a major fault has been the obligatory foundation of case identification on diagnostic criteria, lately classification criteria for RA. Those criteria do not differentiate between RF-positive and RF-negative arthritis which seem to have divergent profiles of risk factors. An additional problem in most studies is that the confounding effects of smoking have not been controlled.

A long-term process characterised by the secretion of RF and related autoantibodies, as discussed earlier often precedes the manifestation of arthritis in RA. Hence, the ultimate causes of RA can hardly be clarified by studying the pathogenesis of arthritis alone. Longitudinal studies focusing on risk factors are likely to be crucial in approaching the aetiology.

At present, information gained from epidemiological studies is vague. Smoking is the only environmental risk factor for RA that has been epidemiologically verified. It can be reasonably regarded as a contributory cause of RA. Studying the immunological effects of exposure to cigarette smoke may offer an opportunity to combine information from basic and epidemiological research to understand the causal chains leading to RA.

In aetiological research on coronary heart disease, the information gained on risk factors and molecular biology today can be meaningfully synthesised. In the case of RA, the ongoing population studies will certainly sharpen views on the alleged risk factors. But combining these data with the knowledge of pathogenesis into an explicit concept would seem to be out of reach in the near future. To bridge the gap, simultaneous orientation in both research areas might create useful innovations.

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